Multiple Cellular Forms of Corticotrophs in Surgically Removed Pituitary Adenomas and Periadenomatous Tissue in Cushing's Disease

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Transsphenoidally removed samples of pituitary adenomas from 14 patients with Cushing's disease and 5 patients with Nelson's syndrome always contained groups of uniform small ACTH-cells. Antibodies against the pro-opiocortin precursor fragments β-endorphin, ACTH, and 16k-peptide recognized material in typical adenoma cells. A subpopulation of these cells, varying in number from sample to sample, specifically exhibited α-melanotropin immunoreactivity. Most periadenomatous samples showed signs of severe degeneration. Typical Crooke cells only occurred in samples from patients with Cushing's syndrome, but, with this exception, no clear differences between pituitaries of patients with Cushing's and Nelson's syndromes could be discerned. Two other forms of ACTH-immunoreactive cells were observed: rare, single, highly immunoreactive cells, with characteristics of both normal and Crooke cells, and numerous syncytial groups of cells in an advanced state of disintegration, presumably the remnants of hyperplastic follicles. The four different corticotrophs are characterized according to their fine structure and immunoreactivity in this study. (Am J Pathol 1982, 106:332-341)

WHEN IT BECAME known that pituitary adrenocorticotropin (ACTH) is part of a large precursor glycoprotein also containing the amino acid sequences of the peptides β-endorphin, β-lipotropin, α-melanotropin (MSH), and a long N-terminal portion, the 16k-fragment,[1-3] the peptide content of human pituitary adenomas in Cushing's disease was reexamined by immunologic methods. Evidence has been presented that human adenoma cells synthesize the whole length of the pro-opiocortin precursor.[4-7] As a consequence, β-endorphin immunoreactivity (IR) coexists with ACTH IR in adenoma cells,[8-11] the two peptides are secreted concomitantly *in vitro*,[6,7,12] and plasma content in β-endorphin increases together with ACTH content in patients with Cushing's syndrome.[13-16] We confirm some of these findings using antibodies against β-endorphin, ACTH and 16k-fragment. Moreover, specific α-MSH IR was found in subpopulations of cells or in cellular granules in microadenomas of both Cushing and Nelson patients.

Absence of circadian rhythm of ACTH release[17] and poor suppression of plasma ACTH by glucocorticoids[18] may indicate an autonomous behavior of the adenomas. However, ACTH secretion in these patients can be suppressed by large doses of glucocorticoids[19] and stimulated by vasopressin administration[20] and metopiron treatment.[21] The differential behavior of adenoma cells and pituitary ACTH-cells in response to adrenal factors is unclear.[22]

In the present structural and immunocytochemical examination of surgically removed microadenomas and periadenomatous pituitary samples, four different forms of ACTH-cells are described. In the case of Crooke cells,[23] structural features can be correlated with high steroid inhibition.[24] Crooke cells were found in pituitaries of patients with Cushing's syn-

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drome, but not of patients with Nelson's syndrome.[25] In other respects, pathologic alterations in pituitary samples from patients with either syndrome were very similar. This may indicate derangements of non-adrenal origin. A better knowledge of the cellular composition of periadenomatous pituitaries may be of prognostic value.[26]

Materials and Methods

Subjects

Adenomatous and periadenomatous tissue samples from pituitaries of 14 patients with Cushing's syndrome (all women, ages 14 to 60 years) and from 5 patients with Nelson's syndrome (3 women, ages 21, 36, and 40 years; 2 men, ages 36 and 57 years), were immersed in fixatives and processed for immunocytochemical and electron-microscopic study immediately after removal by transsphenoidal surgery. The microadenomas measured on average 5 mm (3–10 mm) in diameter; the less compact periadenomatous samples had approximately equal volumes. About half of these tissue blocks were used for routine examination, the other samples for this study.

Fixation and Tissue Processing

Fixative I was composed of 2% glutaraldehyde in 0.1 M phosphate buffer; Fixative II was Zamboni and De Martino's fluid.[27] Fixation time was usually 24 hours. The tissues were dehydrated in graded ethanol and propylene oxide and embedded in Epon. No osmium was applied.

Immunocytochemistry

The immunocytochemical procedures have been described in detail.[28,29] Semithin (0.5 μ) plastic sections were attached by heat to glass slides and the resin was removed by Na-methoxide.[30] After being rinsed in phosphate-buffered saline (PBS: 0.01 M phosphate buffer, 0.9% NaCl, pH 7.3) and 10 minutes' incubation in swine normal serum (1:30 in PBS), primary antibodies raised in rabbits (Table 1) were applied overnight. The antibodies were visualized according to Sternberger et al[31] by the following incubation steps: swine anti-rabbit immunoglobulins (from Dako, Copenhagen, Denmark), 1:100 in PBS, for 10 minutes; a short rinse in PBS; peroxidase-anti-peroxidase complex (from Dako), 1:50 in PBS, for 10 minutes; a rinse in Tris buffer; 3,3'-diaminobenzidine (from Sigma) 20 mg/100 ml Tris buffer, pH 7.6, with 250 μl H_2O_2 1%/100 ml, for 10 minutes in the dark; a rinse in H_2O; dehydration and inclusion in Depex. The sections were examined and photographed with phase contrast optics and green filter, without counterstaining. For localization of different antigens in the same cell, serial 0.5-μ-thick sections were incubated, each with a different antibody. For fine-structural examination serial ultrathin/semithin sections were cut: semithin sections were processed for immunocytochemical study at the light-microscopic level, while ultrathin sections were contrasted with uranyl-

Table 1—List of Antibodies and Their Specificities

Antibodies against	ACTH 11-24*	ACTH 1-39	α-MSH	β-endorphin*	16k-fragment*	Human growth hormone	Met-enkephalin
Purification by affinity chromatography	+	+	+	+	–	–	+
Working dilutions	1:100	1:2000	1:8000	1:1000	1:2000	1:1000	1:2000
Recognized antigens	ACTH 1-39 ACTH 1-24 pro-opiocortin	ACTH 1-24 ACTH 1-28 ACTH 1-39	ACTH 1-13 α-MSH	β-endorphin α-endorphin β-LPH pro-opiocortin	pro-opiocortin 16k-fragment	hGH	Met-enkephalin
Antigens not recognized	β-endorphin Met-enkephalin β-LPH ACTH 25-39 ACTH 19-39 α-MSH 16k-fragment	α-MSH ACTH 4-10 ACTH 25-39 ACTH 19-39 β-endorphin β-LPH	ACTH 1-10 ACTH 4-10 ACTH 1-39 ACTH 18-39 β-MSH	enkephalins ACTH 1-39 α-MSH	α-MSH ACTH 1-39 β-MSH β-endorphin β-LPH Met-enkephalin	β-endorphin ACTH 1-39	β-endorphin Leu-enkephalin

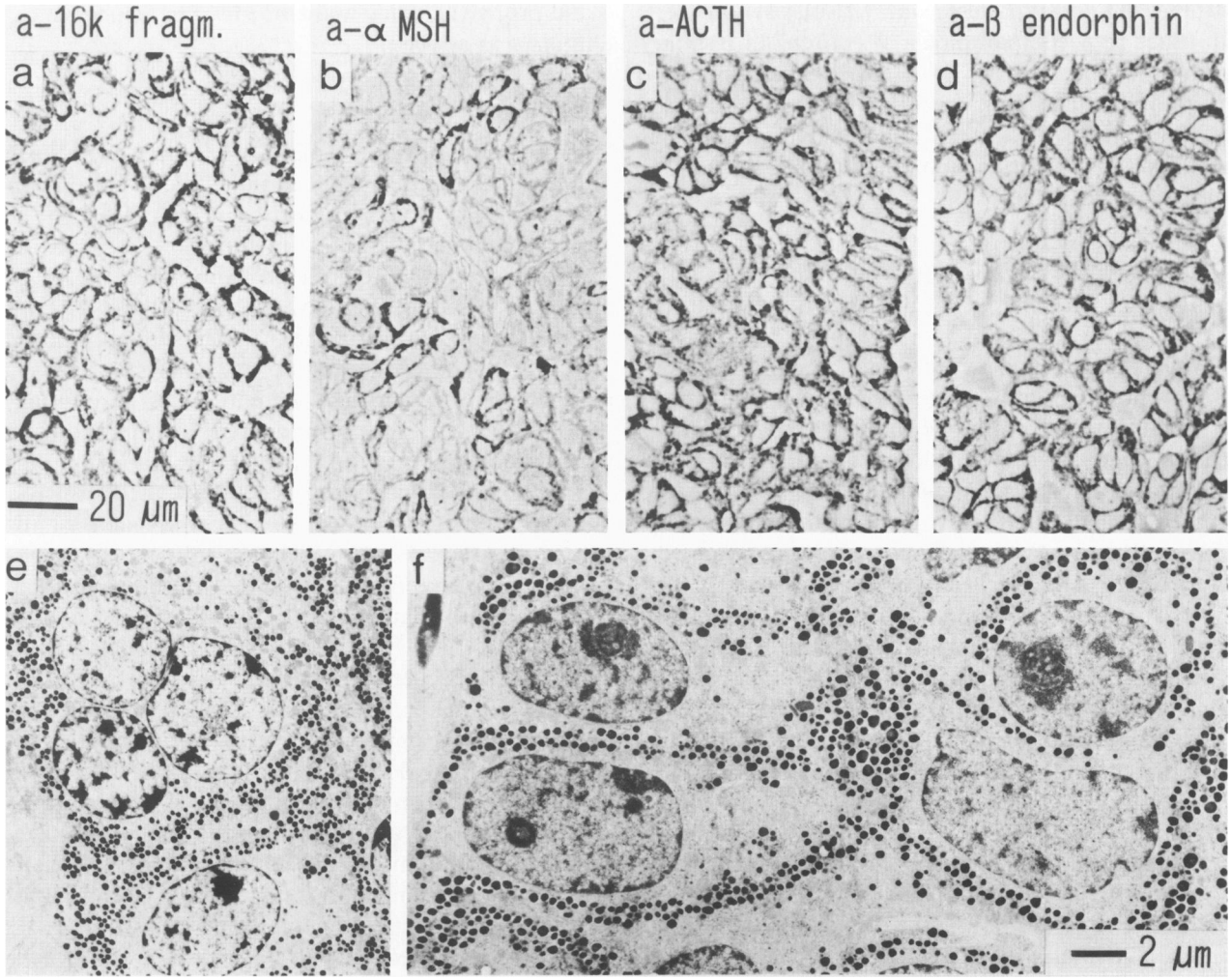


Figure 1 — Light (**a–d**) and electron micrographs (**e, f**) from a microadenoma of a patient with Nelson's syndrome (woman, 40 years old). **a–d** — 0.5-μ-thick sections have been incubated with antibodies against 16k-fragment (**a**), α-MSH (**b**), ACTH (**c**), and β-endorphin (**d**). (×500) **e** — A polynuclear cell that on a consecutive section exhibited α-MSH IR. (×3000) **f** — Fine structure of ACTH immunoreactive adenoma cells. No immunostaining; no osmium, but uranyl acetate and lead citrate contrasting. (×4000) (With a photographic reduction of 5%)

acetate (2% aqueous, for 15 minutes) and lead citrate, for electron-microscopic study.

Specificities of Antibodies and Controls (Table 1)

Antibodies were obtained from Drs. B. A. Eipper and R. E. Mains, of Denver, Colorado (* in Table 1), or were raised by Dr. Voigt. The same antibodies have been used in many studies.[1,28,29,33] They were usually purified by affinity chromatography and were characterized under the conditions of radioimmunoassay. Their specificity was examined in liquid phase preabsorption controls, by incubating for 24 hours 40 μl of antibody solution at working dilution with 5 μg of different peptides (Table 1). Subsequently, the antibody-peptide mixtures were applied on adenoma cell sections. Absence of immunostaining in sections indicated that the antibody had become blocked by the tested peptide; presence of immunostaining that the antibody had not recognized the added peptide. For detection of endogenous peroxidase activity primary antibodies were omitted in the incubation schedule. As a further control, antibodies against human growth hormone and against enkephalin were applied in series with antibodies against pro-opiocortin fragments; human growth hormone-like or Met-enkephalin-like IR was observed in ACTH-cells.

Results

Structure and Immunoreactivity of Microadenomas

A microadenoma of a patient with Nelson's syndrome, a woman, 40 years old, was composed of uniform small cells. Most cells exhibited ACTH, β-endorphin, and 16k-fragment IR (Figure 1a, c, and d).

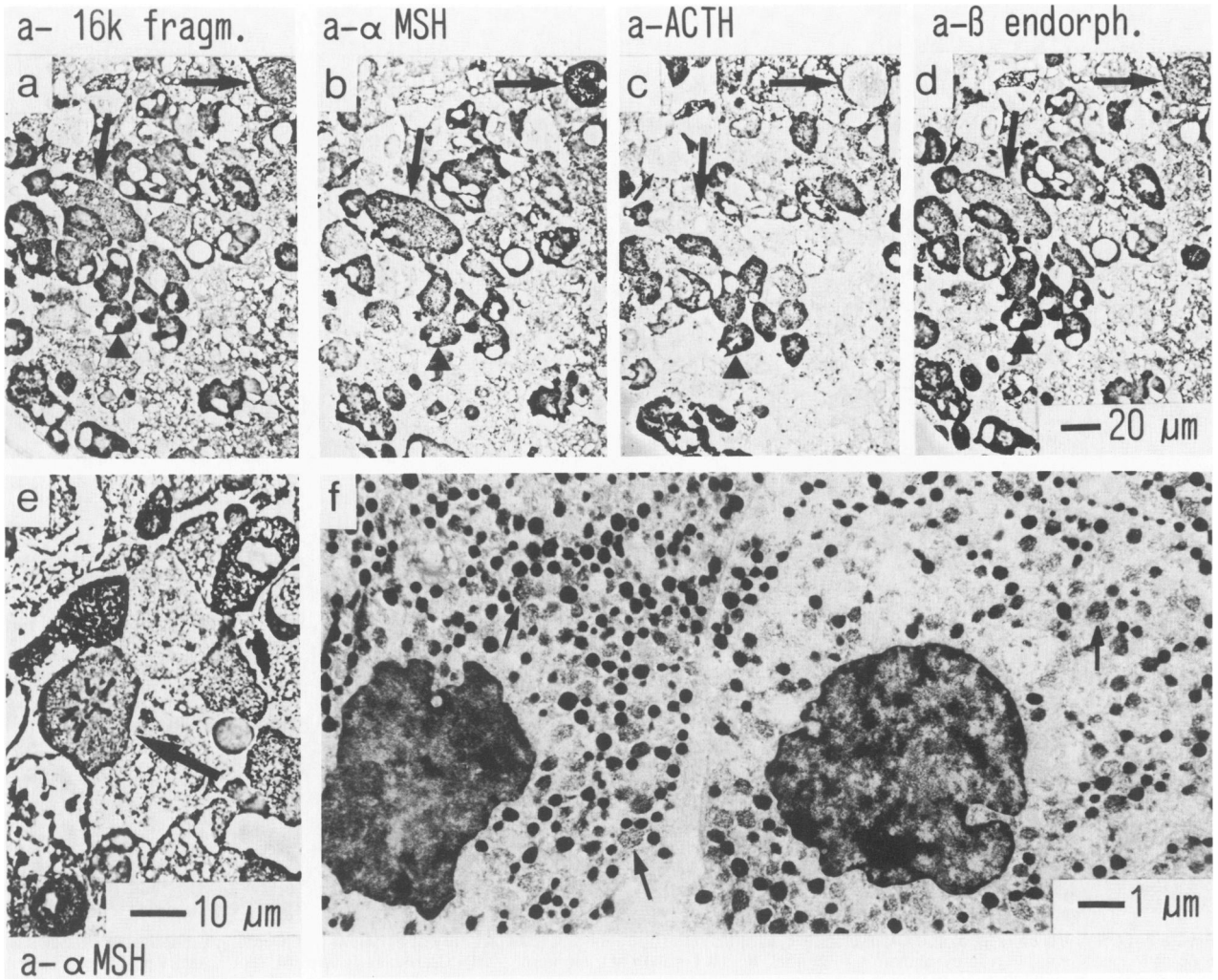


Figure 2—Light (**a–e**) and an electron micrograph (**f**) from a microadenoma of a patient with Nelson's syndrome (male, 56 years old). The serial sections exhibit 16k-fragment, α-MSH, ACTH, and β-endorphin IR; *arrows* point to single cells with α-MSH, but without ACTH IR. (× 300) Section **e** shows an α-MSH immunoreactive cell in mitotic division (*arrow*). (× 800) **f**—Fine structure of adenoma cells that on a subsequent semithin section exhibited ACTH IR. (× 7000) (With a photographic reduction of 5%)

A subpopulation of about 20% of these cells also showed distinct α-MSH IR (Figure 1b). Immunoreactivities were localized in the peripheral area of the cytoplasm, in which, typical for these adenoma cells, the small hormonal granules (276 ± 65 nm in diameter) occurred in rows (Figure 1f). α-MSH immunoreactive cells were not distinctly different in structure, but some were found to contain two or three nuclei (Figure 1e).

In an adenoma from another patient with Nelson's syndrome, a woman, 36 years old, with numerous uniform cells exhibiting ACTH, β-endorphin, and 16k-fragment IR, cells with α-MSH IR were very rare and in many areas absent. An adenoma from a patient with Cushing's syndrome contained a subpopulation of α-MSH cells and cells with a few single large α-MSH immunoreactive granules.

All microadenomas from the 14 patients with Cushing's disease and from 3 of the patients with Nelson's syndrome (all women) contained, at least in some area, clusters of uniform cells with marginal granules that were immunoreactive with antibodies against different pro-opiocortin fragments. Adenoma cells were in close contact with each other, often connected by desmosomes. They surrounded erythrocyte-filled sinuses or touched the walls of blood capillaries with apparently closed endothelium that lacked fenestrae.

A structurally different adenoma had been removed from one of the two male patients with Nelson's syndrome (Figure 2). Wide areas consisted of connective tissue, thick collagen bundles, and fibroblasts. ACTH cells were arranged in loose groups. They were larger than other adenoma cells and less

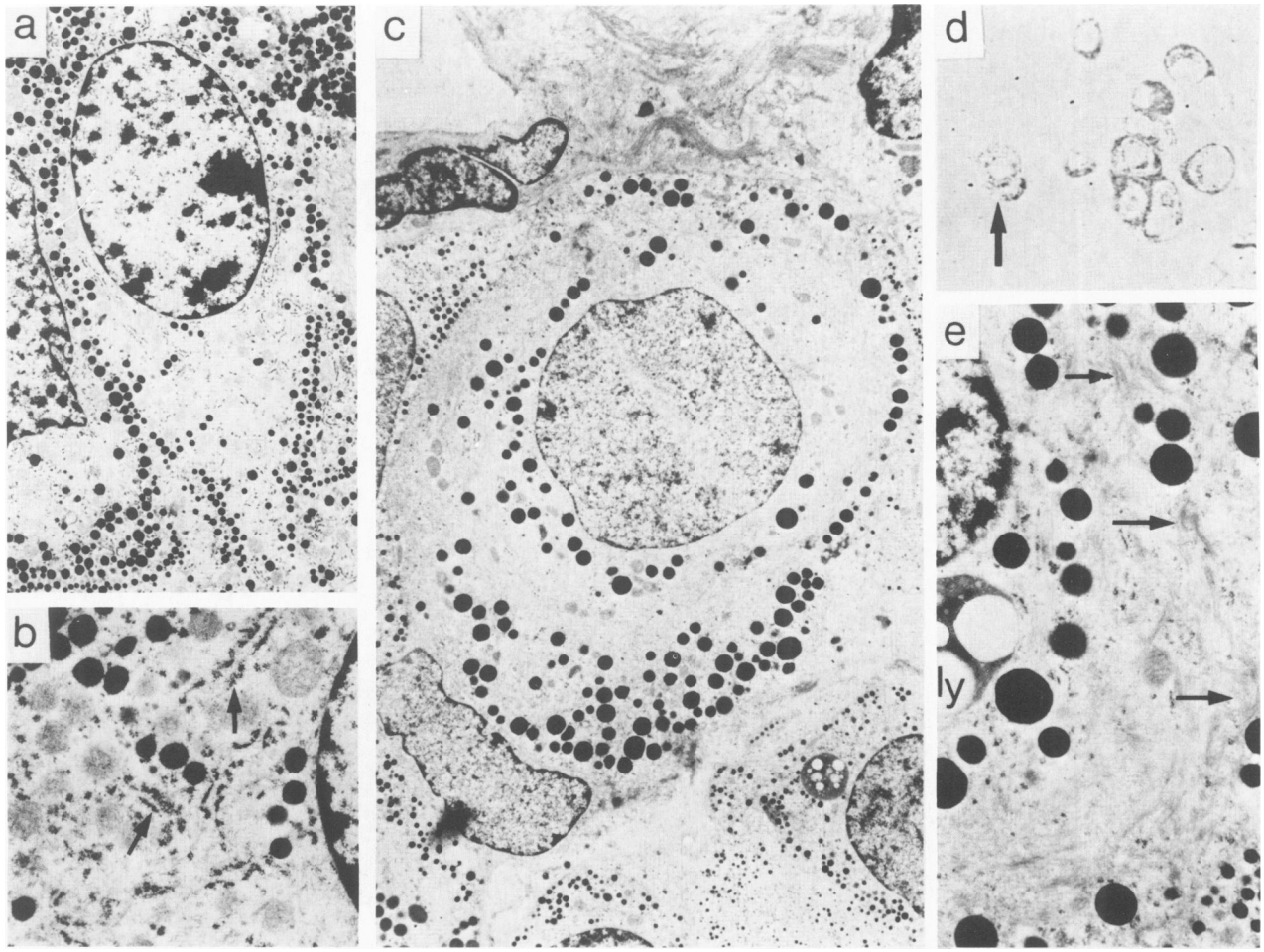


Figure 3 — Adenoma cells (**a, b**) and Crooke cells (**c, d, e**) from the periadenomatous pituitary of a patient with Cushing's disease. The *arrows* in **b** point to ribosomes, in **e** to cytoplasmic filaments. *ly* is a lysosomal inclusion. **d** — Crooke cells with ACTH IR. *Arrow* points to the cell shown in **c**. (**a, c**, × 4500; **b**, × 16,000; **d**, × 450; **e**, × 13,000) (With a photographic reduction of 5%)

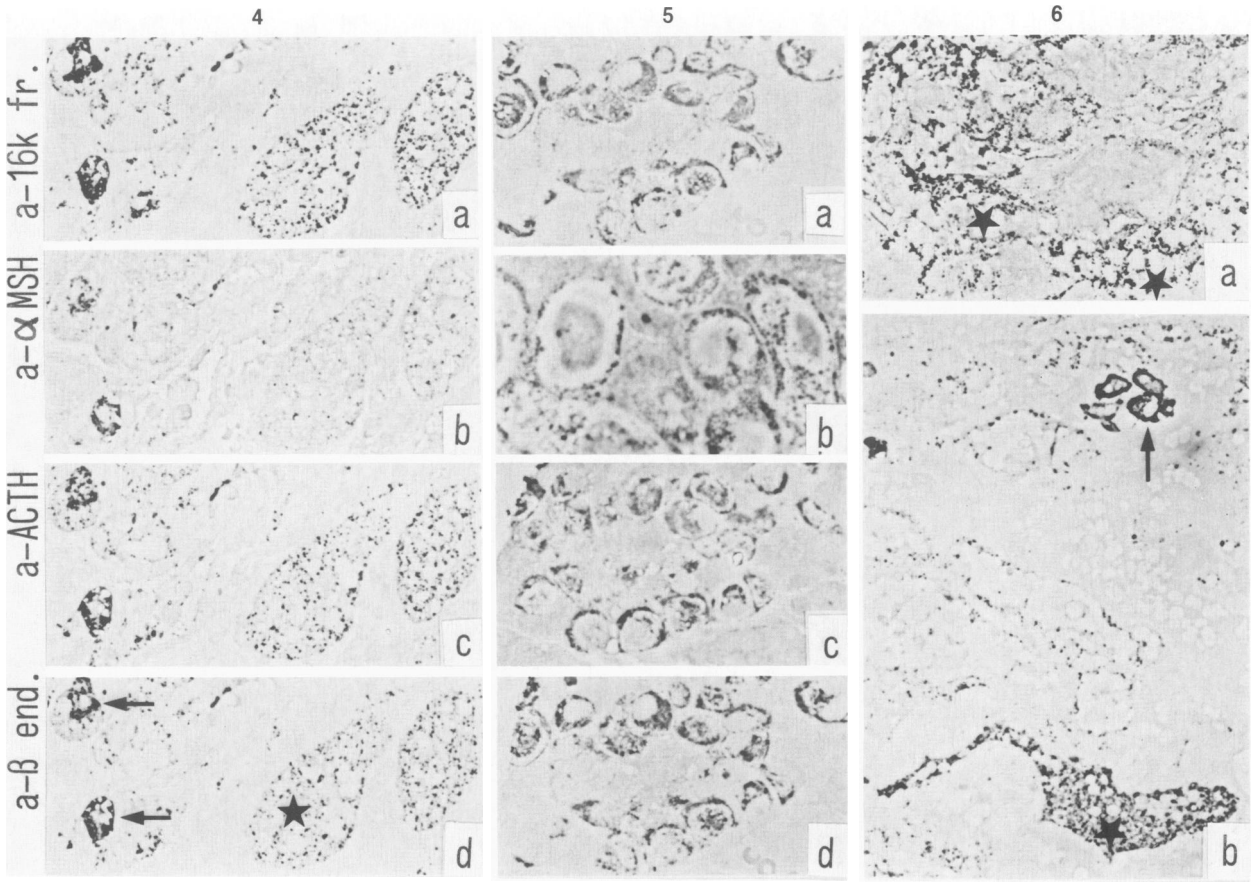
uniform in size and shape. Cells with distinct α-MSH IR, some β-endorphin IR, and 16k-fragment IR, but no ACTH IR, were observed. Other cells were immunoreactive after use of all four antibodies, conserving some ACTH IR besides α-MSH IR (Figure 2a, b, c, and d). Dense distribution of the numerous hormonal granules over the entire cytoplasm rendered these cells intensively immunoreactive (Figure 2f). Microadenomas of the other male patient with Nelson's syndrome similarly showed masses of connective tissue and highly immunoreactive ACTH cells.

Fine Structure and Immunoreactivity of Crooke Cells

Typical Crooke cells were observed in periadenomatous tissue in about half the Cushing cases and in none of the 5 Nelson cases. Crooke cells were much larger than adenoma cells (Figure 3a and c). Also, the hormonal granules had a larger average diameter than the granules of adenoma cells (360 ± 60 nm versus 276 ± 65 nm). Granules were often arranged in two circles: a perinuclear circle and a peripheral circle (Figure 3c and d). The clear area between these circles contained some mitochondria and very few ribosomes, but extensive bundles of fine filaments (Figure 3e). Lysosome-like inclusions were frequent (Figure 3e). Crooke cells exhibited 16k-fragment, α-MSH, ACTH, and β-endorphin IR (Figure 4). The much smaller adenoma cells differed from Crooke cells by a large number of ribosomes and mitochondria, by absence of microfilaments and by scarcity of lysosomal inclusions.

Corticotrophs in Periadenomatous Tissue

Besides Crooke cells, two other forms of corticotrophs were regularly found in samples from both Cushing and Nelson patients: 1) single, highly immunoreactive cells (Figures 4a–d; 6a), hereafter called



Figures 4-6 – Light micrographs of periadenomatous tissue with immunostaining. **Figure 4** – From a patient with Nelson's syndrome (woman, 36 years old). Syncytial follicles (*) and dense cells (*arrows*) after application of antibodies against 16k-fragment (**a**), α-MSH (**b**), ACTH (**c**), and β-endorphin (**d**). (× 300) **Figure 5** – Crooke cells from a patient with Cushing's disease (woman, 15 years old). Immunoreactivities as in Figure 4. (× 300; **5c**, × 1200) **Figure 6** – Periadenomatous samples from two patients with Cushing's disease (women, 56 and 60 years old). ACTH immunoreactive syncytial follicles (*), dense cells (*arrow*), and scattered granules. (× 300) (With a photographic reduction of 9%)

"dense cells"; and 2) large follicular groups of cells with loosely distributed ACTH-immunoreactive granules and many nuclear fragments (Figures 4a–d; 6a; 8), hereafter called "syncytial follicles." Both dense cells and syncytial follicles exhibited 16k-fragment, ACTH, β-endorphin, and some α-MSH IR (Figure 4).

The single, intensely immunoreactive dense cells resembled normal human ACTH cells most closely (Figures 4, 6, and 7). They contained large vacuoles (Figure 7a), like those described in normal corticotrophs.[35] The numerous hormonal granules were spread over the whole cytoplasm (Figure 7a). In Zamboni-fixed samples many of the granules were decomposed and their content was scattered in the cytoplasm (Figure 7c and d). These lytic granules occurred specifically in corticotrophs; granules from neighboring cells were not affected (Figure 7a and d). Dense cells also contained bundles of filaments (Figure 7c). In our periadenomatous tissue samples dense cells were much less frequent than corticotrophs in normal human pituitaries.

Syncytial follicles resembled hyperplastic cell accumulations. They exhibited signs of advanced disintegration (Figure 8). Plasma membranes had disappeared, the less numerous granules were distributed irregularly over wide areas, and there were many lysosomal inclusions. The follicular material was contained in large bags of connective tissue. Typically, syncytial follicles contained many nuclei and nuclear fragments of different sizes. In late stages of degeneration the nuclear material appeared to have shrunken and was very opaque. In periadenomatous samples from both Cushing and Nelson patients syncytial follicles were most frequent and covered large areas of the sections (Figures 4 and 6).

The contrast between the well-preserved microadenomas and the severely damaged periadenomatous pituitary samples was striking. Some of this damage – for example, dispersed erythrocytes or hormonal

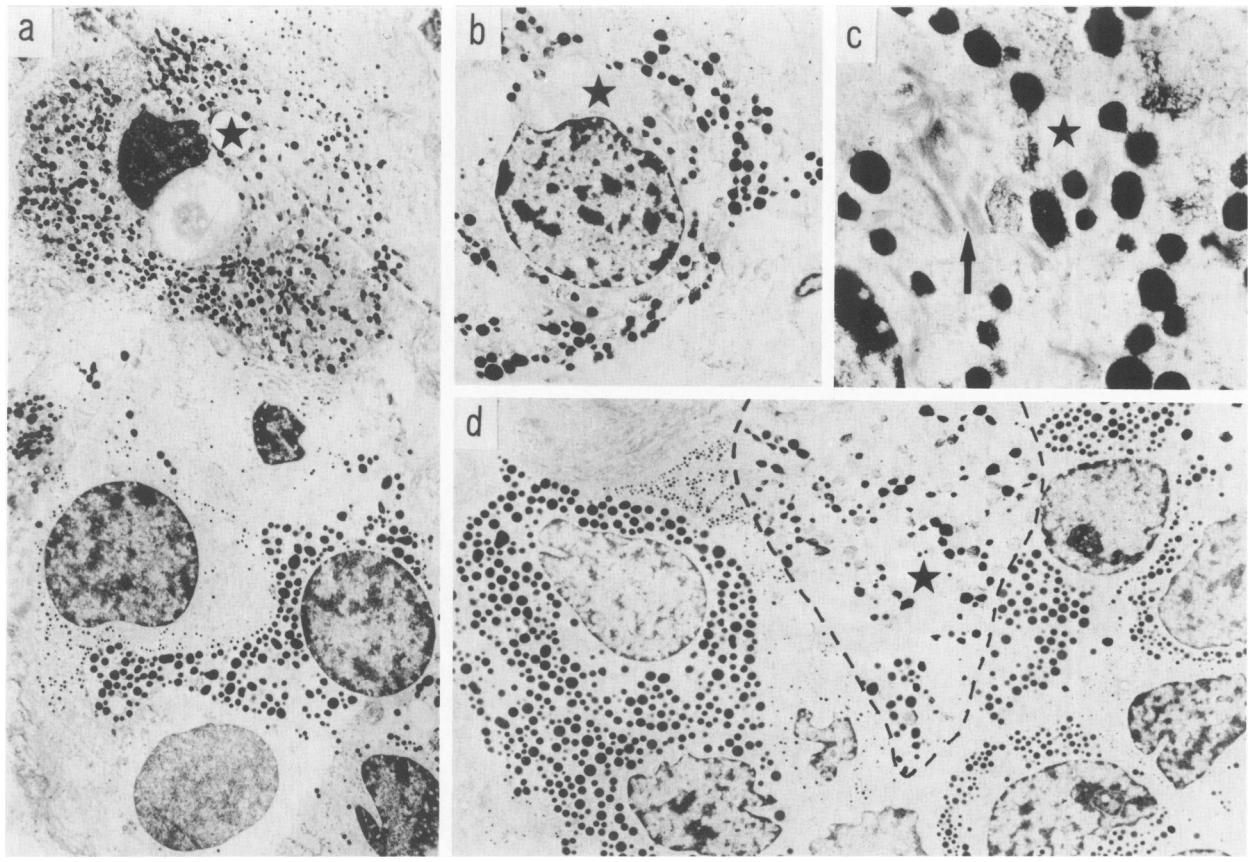


Figure 7 — Electron micrographs of dense cells (*) and other parenchymatous cells in periadenomatous samples from patients with Cushing's syndrome (**a**, woman, 56 years) and Nelson's syndrome (**b, c, d**, woman, 36 years). In ACTH cells (*), but not in other cells, the cytoplasm contains diffuse granular material (**c**). Dense cells also contain filament bundles (*arrow* in **c**). (**a, d**, × 3500; **b**, × 4500; **c**, × 15,000) (With a photographic reduction of 8%)

granules and large empty spaces—was presumably due to handling of the tissue during surgery. However, degenerative changes such as expansion and invasion of connective tissue and fragmentation of nuclei must have developed long before operation. Also, in other hormonal cells, such as thyrotrophs or somatotrophs, the number of lysosomal inclusions was remarkable.

Discussion

The pituitary adenomas removed from 17 female patients with Cushing's or Nelson's disease contained, without exception, groups of small, uniform ACTH cells with the characteristics of actively synthesizing cells. No structural differences were noted between adenoma cells from patients with Cushing's disease and those with Nelson's disease. In contrast to these cells, the adenoma cells from the two male patients were distinctly different.

Antibodies that were raised against the pro-opiocortin precursor fragments β-endorphin, ACTH, and 16k-fragment recognized antigenic material in most of the typical adenoma cells. Since these antibodies do not distinguish between the isolated sequence against which they were raised and this sequence as a portion of the precursor molecule, we do not know to what extent the peptides are stored in the cells as cleaved portions of the precursor. Biochemical studies indicate, however, that ACTH and β-endorphin are secreted by adenoma cells in their specific forms.[12,32]

The applied α-MSH-antibody, however, recognized specifically the C-terminal portion of α-MSH, under radioimmunoassay (RIA) conditions and in preabsorption tests.[33] As in ACTH cells of the normal human pituitary,[34-36] a subpopulation of the adenoma cells exhibited distinct α-MSH IR. The number of α-MSH cells varied from sample to sample: in one adenoma 20% of the cells and in another only a very

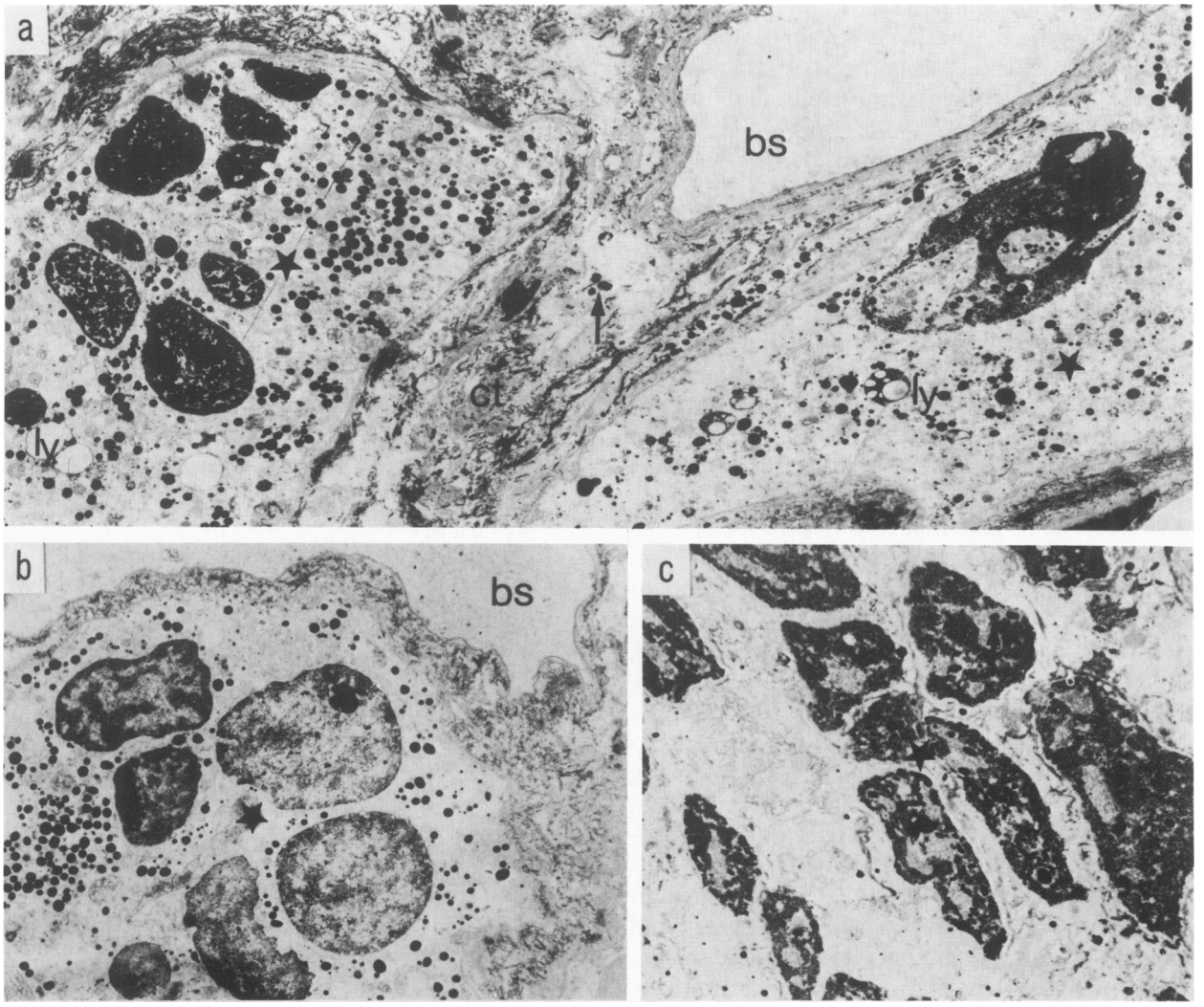


Figure 8 — Advanced degeneration in periadenomatous samples from patients with Nelson's disease (**a**, woman, 40 years old) and Cushing's disease (**b, c**, woman, 56 years old). Syncytial follicles (*) with nuclear fragments and lysosomes (*ly*); dispersed granules (*arrows*), expanding connective tissue (*ct*) and large blood sinuses (*bs*). The syncytial follicles on subsequent semithin sections exhibited ACTH IR. (× 4000) (With a photographic reduction of 8%)

few single cells exhibited immunostaining. In serial sections distinct cells usually exhibited both α-MSH and ACTH IR, but in some α-MSH cells (Figure 2) ACTH IR was absent. These differences are presumably due to partial or complete conversion of ACTH into α-MSH and the corticotropin-like intermediate lobe peptide (CLIP).[37] Melanotropin sequences in this disease are likely to be produced and secreted in varying, often considerable amounts, although they may be absent in single cases.

The main difficulty with the examination of periadenomatous samples is posed by the fact that we do not know how representative these tissue pieces are, in respect to the whole adenohypophyses of these patients. Only samples surrounding the microadenomas could be resected and examined. Therefore, the validity of quantitative statements is limited to small areas. We found that the great majority of the periadenomatous samples exhibited severe degenerative alterations. Changes that must have developed some time before surgical removal consisted of masses of connective tissue, large blood sinuses, or accumulations of lysosomal inclusions in different types of hormonal cells.

In periadenomatous tissue three different forms of ACTH cells were regularly observed. By structural criteria all three cellular forms were pathologic. Crooke cells had typical wide hyaline areas occupied

by bundles of microfilaments, the number of ribosomes was reduced, and large lysosomes were present. Since Crooke cells have not been found after adrenalectomy (Nelson's syndrome), inhibition by high steroid levels due to ACTH overproduction by adenoma cells has been suggested as being responsible for the appearance of these cells.[24,25] In our material, typical Crooke cells were absent in samples from patients who had been subjected to adrenalectomy.

Dense cells most closely resembled normal pituitary ACTH cells.[34-36] However, they were far less frequent than corticotrophs in normal pituitaries. Disintegration of granules, when treated with Zamboni's fixative, may indicate decomposition of granular membranes, and the occurrence of microfilaments in the cytoplasm is a characteristic that dense cells share with Crooke cells.

The very large and numerous syncytial follicles were the most characteristic form of ACTH cell in periadenomatous tissue samples. Accumulation of nuclear fragments indicated that these follicles were degenerated residues of hyperplastic, proliferating cells. At an advanced state of disintegration, hormonal granules were almost completely lost, and necrotic nuclear fragments were grouped together. Extensive impairment of corticotrophs by inhibiting factors or by degeneration correlates well with observed low plasma ACTH-levels over extended periods after surgical removal of microadenomas.[9]

With the exception of typical Crooke cells, pituitary tissues from patients with Cushing's disease and those with Nelson's syndrome were indistinguishable. In both kinds of samples, tissue disintegration prevailed, and dense cells and syncytial follicles were present. Thus, factors not necessarily of adrenal origin may be responsible for pathologic changes in pituitary corticotrophs.

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